

Exercise

Give tableau proofs for the following logical consequences:

- $\forall x.P(x) \vee \forall x.Q(x) \models \neg \exists x(\neg P(x) \wedge \neg Q(x))$
- $\models \exists x.(P(x) \vee Q(x)) \equiv \exists x.P(x) \vee \exists x.Q(x)$

Some definition for tableaux

Definition (Closed branch)

A **closed branch** is a branch which contains a formula and its negation.

Definition (Open branch)

An **open branch** is a branch which is not closed

Definition (Closed tableaux)

A tableaux is **closed** if all its branches are closed.

Definition

Let ϕ be a first-order formula and Γ a finite set of such formulas. We write $\Gamma \vdash \phi$ to say that there exists a closed tableau for $\Gamma \cup \{\neg\phi\}$

Soundness and completeness

Theorem (Soundness)

$$\Gamma \vdash \phi \quad \Longrightarrow \quad \Gamma \models \phi$$

Theorem (Completeness)

$$\Gamma \models \phi \quad \Longrightarrow \quad \Gamma \vdash \phi$$

Remark

The mere existence of a closed tableau does not mean that we have an effective method to build it! Concretely: we don't know how often and in which way we have to apply] the γ -rules ($\forall x\phi(x) \Rightarrow \phi[x/t]$), and what term to use in the substitution.

Example

Check via tableaux if the validity/satisfiability of the formula
 $\phi = \forall x, y(P(x) \supset Q(y)) \supset (\exists x P(x) \supset \forall y Q(y))$

Solution

$\neg(\forall xy(P(x) \supset Q(y)) \supset (\exists x P(x) \supset \forall y Q(y)))$

$\forall xy(P(x) \supset Q(y))$
 $\neg(\exists x P(x) \supset \forall y Q(y))$

$\exists x P(x)$
 $\neg \forall y Q(y)$

$P(a)$

$\neg Q(b)$

$P(a) \supset Q(b)$

$\neg P(a)$

$Q(b)$

CLASH

CLASH

We try, with the tableaux, to build a model for the negation of ϕ . Since the tableaux ends with all CLASHES, there is no such a model. In other words, for all $\mathcal{I}$, $\mathcal{I} \not\models \neg\phi$. Which implies that for all $\mathcal{I}$, $\mathcal{I} \models \phi$, i.e., that ϕ is valid.

Infinite domains

- Differently from Prop. Logic, in FOL, models can be infinite.
- There are formulas which are satisfied only by infinite models. For instance the following formula¹

$$\phi = \left(\begin{array}{c} \forall x \neg R(x, x) \quad \wedge \\ \forall xyz. (R(x, y) \wedge R(y, z) \supset R(x, z)) \quad \wedge \\ \forall x. \exists y. R(x, y) \end{array} \right)$$

- If we build a tableaux for such a formula, searching for a model, we will end up in an infinite tableaux.

¹To verify this, suppose that $\mathcal{I} = \langle \Delta^{\mathcal{I}}, \cdot^{\mathcal{I}} \rangle$ is an interpretation that satisfies ϕ , and suppose that $|\Delta| = n$ for some finite number n . Consider the sequence $\langle d_1, d_2, d_3, \dots, d_{n+1} \rangle$ of $n+1$ elements of Δ , such that $\langle d_i, d_{i+1} \rangle \in R^{\mathcal{I}}$. This sequence exists, because for every d there is always a d' with $\langle d, d' \rangle \in R^{\mathcal{I}}$, since $\mathcal{I} \models \forall x. \exists y. R(x, y)$. $\mathcal{I} \models \forall xyz. (R(x, y) \wedge R(y, z) \supset R(x, z))$, implies that $R^{\mathcal{I}}$ is transitive, and therefore for all $0 \leq i < j \leq n+1$, $\langle d_i, d_j \rangle \in R^{\mathcal{I}}$. The fact that Δ contains at most n elements implies that for some $1 \leq i < j \leq n+1$, $d_i = d_j$, which means that $\langle d_i, d_i \rangle \in R^{\mathcal{I}}$ for some $1 \leq i \leq n$. But this contradicts the fact that $\mathcal{I} \models \forall x \neg R(x, x)$.

Infinite tableaux

Exercise

Build a tableaux for

$$\forall x \neg R(x, x) \wedge \forall xyz. (R(x, y) \wedge R(y, z) \supset R(x, z)) \wedge \forall x. \exists y. R(x, y)$$

Solution

$$\forall x \neg R(x, x) \wedge \forall xyz. (R(x, y) \wedge R(y, z) \supset R(x, z)) \wedge \forall x. \exists y. R(x, y)$$

$$\begin{array}{c} | \\ \forall x \neg R(x, x) \\ \forall xyz. (R(x, y) \wedge R(y, z) \supset R(x, z)) \\ \forall x. \exists y. R(x, y) \end{array}$$

$$\begin{array}{c} | \\ \exists y. R(a_0, y) \end{array}$$

$$\begin{array}{c} | \\ R(a_0, a_1) \end{array}$$

$$\begin{array}{c} | \\ \exists y. R(a_1, y) \end{array}$$

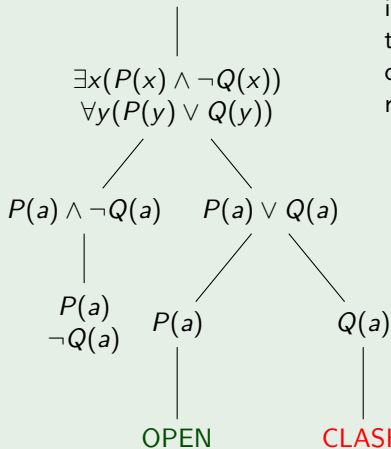
$\vdots$

By applying the γ -rule to the axiom $\forall x \exists y (R(x, y))$, we generate $\exists y R(a_0, y)$ for an initial constant a_0 , and by applying the δ -rule to this last formula we generate a new individual a_1 . This allow to apply the γ -rule again to $\forall x \exists y R(x, y)$, obtaining $\exists y R(a_1, y)$, and again by applying δ -rule to this new formula we generate another constant a_2 . The process can go on infinitely without reaching any clash

Example of tableaux

Example

$\exists x(P(x) \wedge \neg Q(x)) \wedge \forall y(P(y) \vee Q(y))$



From the formulas appearing in the **OPEN** branch of the tableaux it is possible to construct a model for the root formula.

- $\Delta = \{a\}$, the constants appearing in the formulas
- $I(P) = \{a\}$, since the formula $P(a)$ appears in the open branch
- $I(Q) = \{\}$ since the formula $\neg Q(a)$ appears in the open branch

Termination fo FO tableaux

In contrast to what happens in propositional logic, the tableau construction is not guaranteed to terminate.

If the formula ϕ that labels the root is unsatisfiable, in which case the construction is guaranteed to terminate and the tableau is closed.

If the formula ϕ that labels the root is satisfiable then either the construction is guaranteed to terminate and the tableau is open, or the construction does not terminate.

First order logic

Evaluation and reasoning

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First order logic

- First order logic (FOL) is the logic to speak about **objects**, which are the domain of discourse or universe.
- FOL is concerned about **properties** of these objects and **relations** over objects (resp., unary and n -ary **predicates**).
- FOL also has **functions** including **constants** that denote objects.

FOL syntax – Terms

We first introduce:

- A set $Vars = \{x_1, \dots, x_n\}$ of **individual variables** (i.e., variables that denote single objects).
- A set of **functions symbols**, each of given arity ≥ 0 .
Functions of arity 0 are called **constants**.

Def.: The set of **Terms** is defined inductively as follows:

- $Vars \subseteq Terms$;
- If $t_1, \dots, t_k \in Terms$ and f^k is a k -ary function symbol, then $f^k(t_1, \dots, t_k) \in Terms$;
- Nothing else is in $Terms$.

FOL syntax – Formulas

Def.: The set of *Formulas* is defined inductively as follows:

- If $t_1, \dots, t_k \in \text{Terms}$ and P^k is a k -ary predicate, then $P^k(t_1, \dots, t_k) \in \text{Formulas}$ (atomic formulas).
- If $t_1, t_2 \in \text{Terms}$, then $t_1 = t_2 \in \text{Formulas}$.
- If $\varphi \in \text{Formulas}$ and $\psi \in \text{Formulas}$ then
 - $\neg\varphi \in \text{Formulas}$
 - $\varphi \wedge \psi \in \text{Formulas}$
 - $\varphi \vee \psi \in \text{Formulas}$
 - $\varphi \rightarrow \psi \in \text{Formulas}$
- If $\varphi \in \text{Formulas}$ and $x \in \text{Vars}$ then
 - $\exists x.\varphi \in \text{Formulas}$
 - $\forall x.\varphi \in \text{Formulas}$
- Nothing else is in *Formulas*.

Note: a predicate of arity 0 is a proposition of propositional logic.

Interpretations

Given an **alphabet** of predicates $P_1, P_2, \dots$ and functions $f_1, f_2, \dots$, each with an associated arity, a FOL **interpretation** is:

$$\mathcal{I} = (\Delta^{\mathcal{I}}, P_1^{\mathcal{I}}, P_2^{\mathcal{I}}, \dots, f_1^{\mathcal{I}}, f_2^{\mathcal{I}}, \dots)$$

where:

- $\Delta^{\mathcal{I}}$ is the domain (a set of objects)
- if P_i is a k -ary predicate, then $P_i^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \dots \times \Delta^{\mathcal{I}}$ (k times)
- if f_i is a k -ary function, then $f_i^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \dots \times \Delta^{\mathcal{I}} \longrightarrow \Delta^{\mathcal{I}}$ (k times)
- if f_i is a constant (i.e., a 0-ary function), then $f_i^{\mathcal{I}} : () \longrightarrow \Delta^{\mathcal{I}}$
(i.e., f_i denotes exactly one object of the domain)

Assignment

Let $Vars$ be a set of (individual) variables.

Def.: Given an interpretation $\mathcal{I}$, an **assignment** is a function

$$\alpha : Vars \longrightarrow \Delta^{\mathcal{I}}$$

that assigns to each variable $x \in Vars$ an object $\alpha(x) \in \Delta^{\mathcal{I}}$.

It is convenient to extend the notion of assignment to terms. We can do so by defining a function $\hat{\alpha} : Terms \longrightarrow \Delta^{\mathcal{I}}$ inductively as follows:

- $\hat{\alpha}(x) = \alpha(x)$, if $x \in Vars$
- $\hat{\alpha}(f(t_1, \dots, t_k)) = f^{\mathcal{I}}(\hat{\alpha}(t_1), \dots, \hat{\alpha}(t_k))$

Note: for constants $\hat{\alpha}(c) = c^{\mathcal{I}}$.

Truth in an interpretation wrt an assignment

We define when a FOL formula φ is **true** in an interpretation $\mathcal{I}$ wrt an assignment α , written $\mathcal{I}, \alpha \models \varphi$:

- $\mathcal{I}, \alpha \models P(t_1, \dots, t_k)$ if $(\hat{\alpha}(t_1), \dots, \hat{\alpha}(t_k)) \in P^{\mathcal{I}}$
- $\mathcal{I}, \alpha \models t_1 = t_2$ if $\hat{\alpha}(t_1) = \hat{\alpha}(t_2)$
- $\mathcal{I}, \alpha \models \neg\varphi$ if $\mathcal{I}, \alpha \not\models \varphi$
- $\mathcal{I}, \alpha \models \varphi \wedge \psi$ if $\mathcal{I}, \alpha \models \varphi$ and $\mathcal{I}, \alpha \models \psi$
- $\mathcal{I}, \alpha \models \varphi \vee \psi$ if $\mathcal{I}, \alpha \models \varphi$ or $\mathcal{I}, \alpha \models \psi$
- $\mathcal{I}, \alpha \models \varphi \rightarrow \psi$ if $\mathcal{I}, \alpha \models \varphi$ implies $\mathcal{I}, \alpha \models \psi$
- $\mathcal{I}, \alpha \models \exists x.\varphi$ if for some $a \in \Delta^{\mathcal{I}}$ we have $\mathcal{I}, \alpha[x \mapsto a] \models \varphi$
- $\mathcal{I}, \alpha \models \forall x.\varphi$ if for every $a \in \Delta^{\mathcal{I}}$ we have $\mathcal{I}, \alpha[x \mapsto a] \models \varphi$

Here, $\alpha[x \mapsto a]$ stands for the new assignment obtained from α as follows:

$$\begin{aligned}\alpha[x \mapsto a](x) &= a \\ \alpha[x \mapsto a](y) &= \alpha(y) \quad \text{for } y \neq x\end{aligned}$$